

## DESCHAMBAULT, QC, Canada

WMO: 713890

Lat: 46.6906N

Lon: 71.9717W

Elev: 61

StdP: 100.59

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -28.0      | -25.1 | -32.5                         | 0.2 | -27.1 | -29.5 | 0.2 | -24.3 | 8.3                   | -8.4 | 7.4 | -8.3 | 1.2                   | 320  | 0.545 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.1                   | 29.2            | 21.5 | 27.7 | 20.5 | 26.2 | 19.6 | 23.2                | 27.2 | 22.1 | 25.9 | 21.0 | 24.5 | 2.3                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.0                            | 16.8 | 25.7 | 20.8 | 15.6 | 24.0 | 19.8 | 14.6 | 23.0 | 69.4          | 27.0 | 65.0 | 26.0 | 61.2 | 24.5 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.2               | 6.3  | 5.5 | -33.1                      | 32.0 | 2.8                | 1.5 | -35.2                                              | 33.1 | -36.8      | 33.9 | -38.4      | 34.8 | -40.5      | 35.9 |
|                   |      |     | -33.0                      | 25.3 | 2.8                | 1.4 | -35.1                                              | 26.3 | -36.7      | 27.1 | -38.3      | 27.8 | -40.4      | 28.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 4.6         | -11.9 | -10.1 | -4.1  | 3.7   | 11.4  | 16.0  | 18.9  | 17.8  | 13.5  | 6.7   | 0.2   | -7.5  |
|                                                                   | DBStd       | 11.77       | 7.34  | 6.64  | 6.15  | 4.03  | 4.07  | 3.53  | 2.87  | 2.99  | 3.98  | 4.16  | 4.92  | 6.51  |
|                                                                   | HDD10.0     | 2872        | 679   | 562   | 436   | 191   | 32    | 2     | 0     | 0     | 12    | 119   | 296   | 544   |
|                                                                   | HDD18.3     | 5106        | 937   | 795   | 694   | 438   | 219   | 87    | 28    | 46    | 153   | 361   | 544   | 802   |
|                                                                   | CDD10.0     | 908         | 0     | 0     | 0     | 3     | 74    | 181   | 274   | 243   | 116   | 17    | 1     | 0     |
|                                                                   | CDD18.3     | 102         | 0     | 0     | 0     | 0     | 3     | 17    | 44    | 31    | 7     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 1187        | 0     | 0     | 0     | 2     | 81    | 232   | 487   | 300   | 84    | 1     | 0     | 0     |
|                                                                   | CDH26.7     | 247         | 0     | 0     | 0     | 0     | 15    | 53    | 114   | 51    | 14    | 0     | 0     | 0     |
| Wind                                                              | WSAvg       | 2.2         | 2.3   | 2.5   | 2.7   | 2.8   | 2.5   | 2.0   | 1.6   | 1.6   | 1.7   | 2.1   | 2.2   | 2.2   |
| Precipitation                                                     | PrecAvg     | 1223        | 87    | 71    | 73    | 85    | 95    | 113   | 141   | 117   | 123   | 116   | 96    | 103   |
|                                                                   | PrecMax     | 1406        | 141   | 117   | 133   | 194   | 154   | 209   | 239   | 176   | 232   | 234   | 158   | 226   |
|                                                                   | PrecMin     | 989         | 39    | 27    | 20    | 19    | 51    | 50    | 70    | 37    | 62    | 40    | 35    | 46    |
|                                                                   | PrecStd     | 123         | 30    | 22    | 26    | 43    | 29    | 43    | 43    | 30    | 42    | 44    | 28    | 43    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 4.9   | 5.4   | 12.1  | 21.1  | 28.7  | 30.5  | 32.0  | 29.8  | 28.6  | 22.1  | 14.3  | 6.1   |
|                                                                   |             | MCWB        | 4.2   | 2.9   | 6.1   | 12.0  | 18.1  | 21.4  | 24.2  | 22.1  | 22.1  | 16.4  | 11.2  | 4.9   |
|                                                                   | 2%          | DB          | 2.4   | 3.2   | 7.9   | 17.3  | 25.6  | 28.2  | 29.5  | 28.1  | 25.8  | 18.5  | 11.0  | 4.0   |
|                                                                   |             | MCWB        | 1.6   | 1.2   | 3.3   | 9.7   | 16.1  | 19.9  | 22.2  | 21.2  | 20.1  | 14.0  | 8.8   | 3.1   |
|                                                                   | 5%          | DB          | 0.8   | 1.7   | 5.8   | 13.9  | 22.8  | 26.1  | 27.8  | 26.5  | 23.4  | 15.7  | 8.8   | 2.2   |
|                                                                   |             | MCWB        | 0.2   | 0.4   | 2.5   | 7.5   | 14.8  | 18.8  | 21.0  | 20.3  | 18.6  | 12.4  | 6.9   | 1.4   |
|                                                                   | 10%         | DB          | -1.2  | 0.1   | 3.9   | 11.1  | 20.0  | 23.9  | 26.2  | 24.9  | 21.1  | 13.4  | 6.9   | 0.9   |
|                                                                   |             | MCWB        | -2.1  | -0.9  | 1.4   | 5.9   | 13.3  | 17.4  | 20.2  | 19.6  | 17.2  | 10.8  | 5.4   | 0.3   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 4.4   | 3.5   | 7.4   | 14.5  | 21.0  | 23.5  | 25.2  | 24.2  | 23.1  | 17.6  | 12.6  | 5.5   |
|                                                                   |             | MCDB        | 4.7   | 4.8   | 9.7   | 20.1  | 25.8  | 27.7  | 30.0  | 28.4  | 26.7  | 20.3  | 13.6  | 5.8   |
|                                                                   | 2%          | WB          | 1.8   | 1.7   | 4.5   | 10.3  | 18.2  | 21.8  | 23.6  | 22.6  | 21.2  | 14.8  | 9.2   | 3.3   |
|                                                                   |             | MCDB        | 2.2   | 2.7   | 6.8   | 14.8  | 22.6  | 26.1  | 27.7  | 26.2  | 24.2  | 17.3  | 10.2  | 3.9   |
|                                                                   | 5%          | WB          | 0.3   | 0.5   | 3.0   | 8.4   | 16.0  | 20.3  | 22.3  | 21.5  | 19.5  | 13.1  | 7.0   | 1.6   |
|                                                                   |             | MCDB        | 0.8   | 1.5   | 5.0   | 12.8  | 19.7  | 24.1  | 26.2  | 24.8  | 22.1  | 15.1  | 8.3   | 2.1   |
|                                                                   | 10%         | WB          | -1.9  | -0.9  | 1.6   | 6.7   | 14.2  | 18.9  | 21.2  | 20.5  | 17.9  | 11.2  | 5.5   | 0.4   |
|                                                                   |             | MCDB        | -1.3  | 0.2   | 3.6   | 10.1  | 19.1  | 22.5  | 24.8  | 23.6  | 20.5  | 12.9  | 6.7   | 0.9   |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.9  | 12.1  | 11.2  | 11.3  | 13.2  | 13.1  | 13.1  | 13.0  | 13.0  | 10.3  | 8.2   | 8.9   |
|                                                                   |             | MCDBR       | 8.9   | 9.7   | 12.5  | 17.5  | 18.5  | 16.5  | 15.4  | 14.9  | 15.4  | 14.1  | 10.8  | 7.0   |
|                                                                   | 5% WB       | MCWBR       | 8.8   | 8.1   | 8.6   | 10.7  | 10.2  | 9.3   | 8.9   | 8.6   | 9.8   | 9.9   | 8.8   | 6.6   |
|                                                                   |             | MCWBR       | 8.8   | 8.7   | 10.5  | 14.9  | 15.0  | 13.8  | 13.5  | 12.7  | 13.3  | 12.6  | 9.3   | 6.8   |
| Clear-Sky Solar Irradiance                                        | taub        |             | 0.265 | 0.283 | 0.320 | 0.348 | 0.366 | 0.407 | 0.411 | 0.399 | 0.371 | 0.337 | 0.308 | 0.284 |
|                                                                   |             | taud        | 2.309 | 2.245 | 2.244 | 2.372 | 2.401 | 2.283 | 2.288 | 2.326 | 2.395 | 2.490 | 2.481 | 2.320 |
|                                                                   | Ebn at Noon |             | 832   | 887   | 901   | 907   | 899   | 860   | 850   | 846   | 840   | 822   | 781   | 758   |
|                                                                   |             | Edh at Noon | 75    | 98    | 116   | 113   | 114   | 129   | 127   | 117   | 100   | 79    | 65    | 67    |
| All-Sky Solar Radiation                                           | RadAvg      |             | 1.41  | 2.31  | 3.57  | 4.49  | 5.09  | 5.56  | 5.45  | 4.75  | 3.72  | 2.17  | 1.27  | 0.95  |
|                                                                   | RadStd      |             | 0.20  | 0.28  | 0.31  | 0.41  | 0.53  | 0.38  | 0.52  | 0.26  | 0.31  | 0.18  | 0.16  | 0.09  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (39 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page